

Lecture 2: Video Imaging



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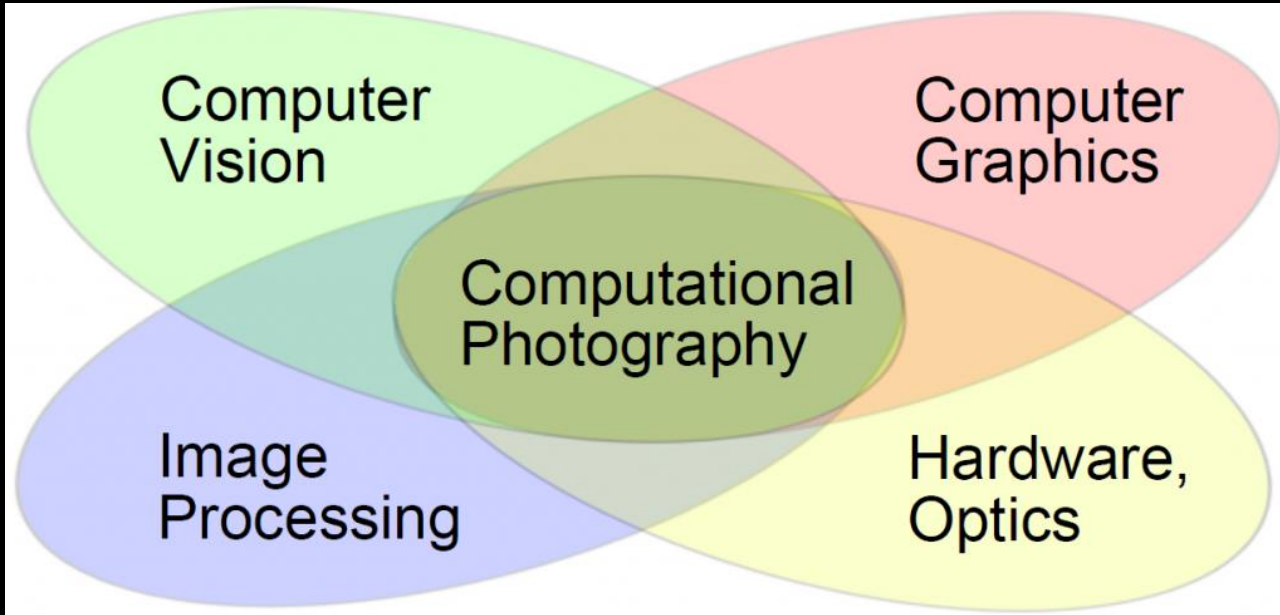
Academic website <https://nelson-diaz.com/>



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FEDERICO SANTA MARIA



Computational Photography

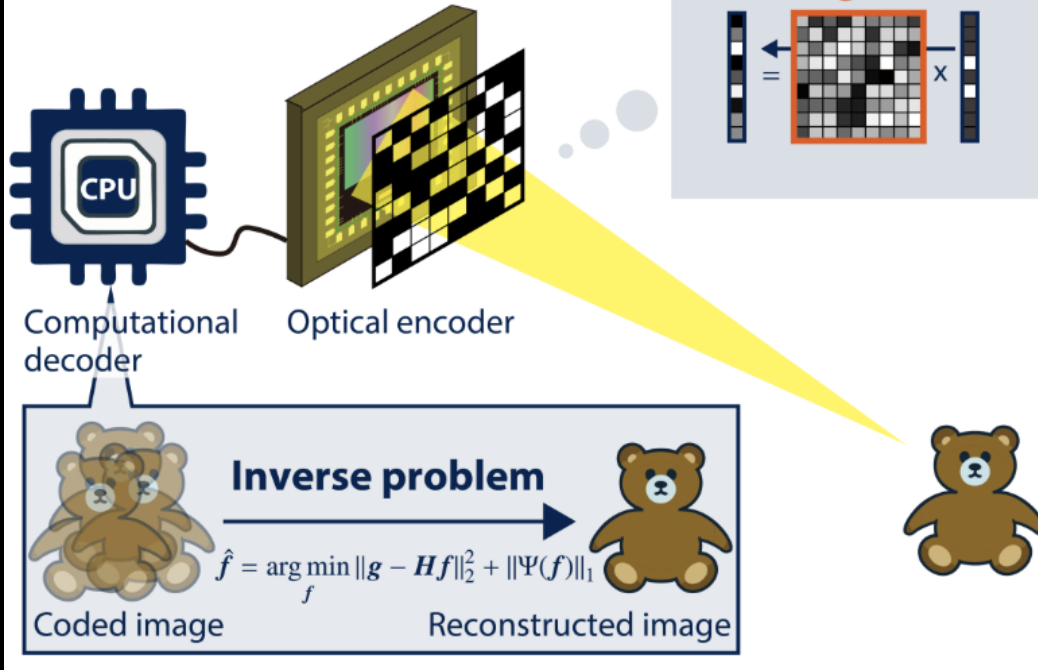


Areas:

- Computer Science
- Electrical Engineering and Electronics
- Maths
- Physics (optics)

Computational Photography

Computational imaging



Goal:

- To capture and recover signal in many dimensions

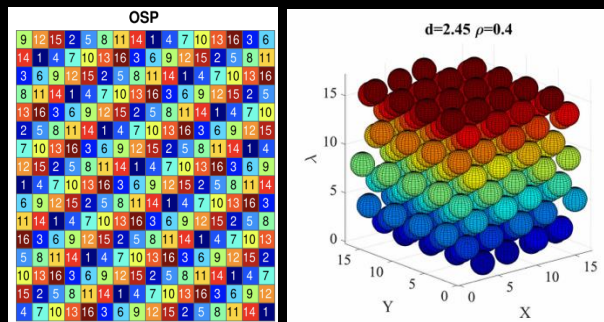
To design

- Hardware
- Software--algorithm to solve the inverse problem--

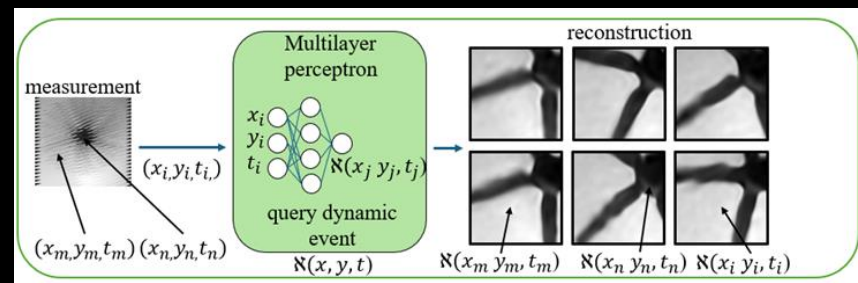
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral



Video

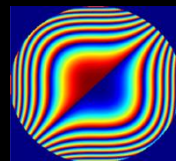
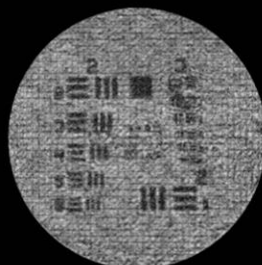


GT phase

Recovered phase

Recovered Amplitude

A woman

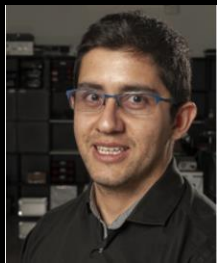


1995
Cubic Phase



Depth

Single-mask sphere-packing with implicit neural representation for ultrahigh-speed imaging

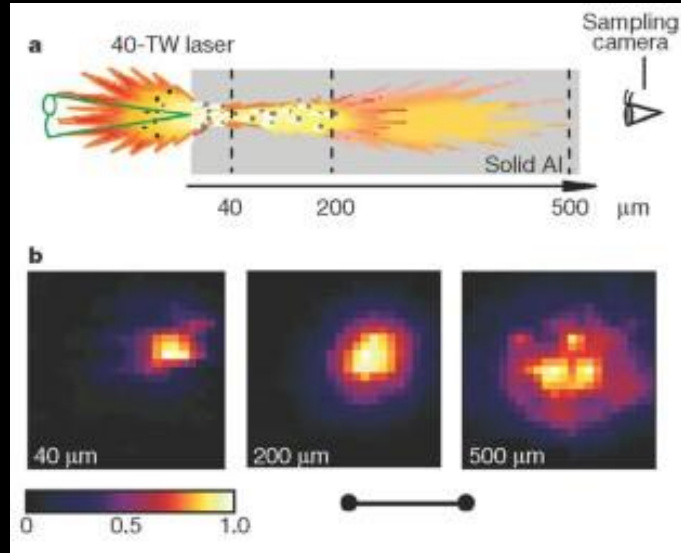


Nelson Diaz¹, Madhu Beniwal², Miguel Marquez², Felipe Guzman¹, Cheng Jiang², Esteban Vera¹, Jinyang Liang²

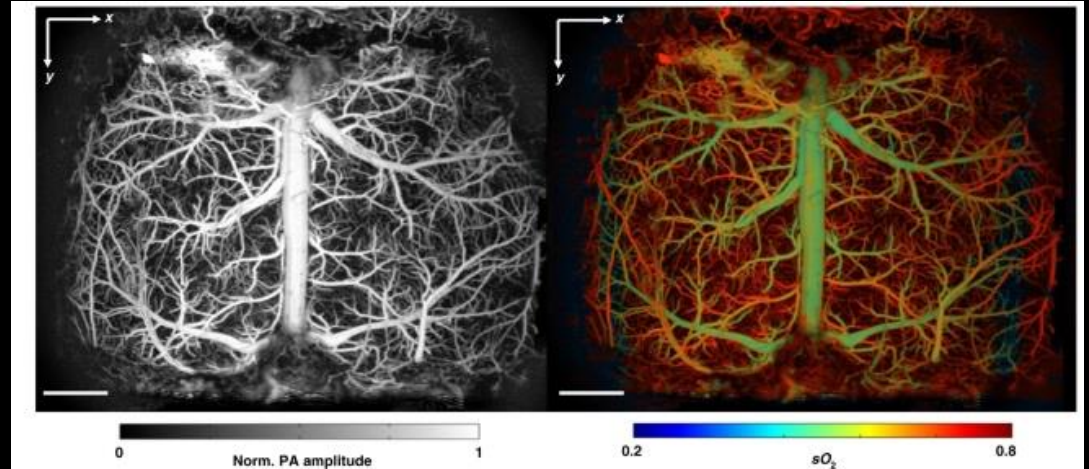
¹ School of Electrical Engineering, Pontificia Universidad Católica de Valparaíso, Valparaíso, Chile

² Laboratory of Applied Computational Imaging, Centre Énergie Matériaux Télécommunications, Institut National de la Recherche Scientifique, Montreal, Canada

Sensing Transient Events



Nuclear fusion: Fast-heating plasma



Neuroscience for monitoring neural activity

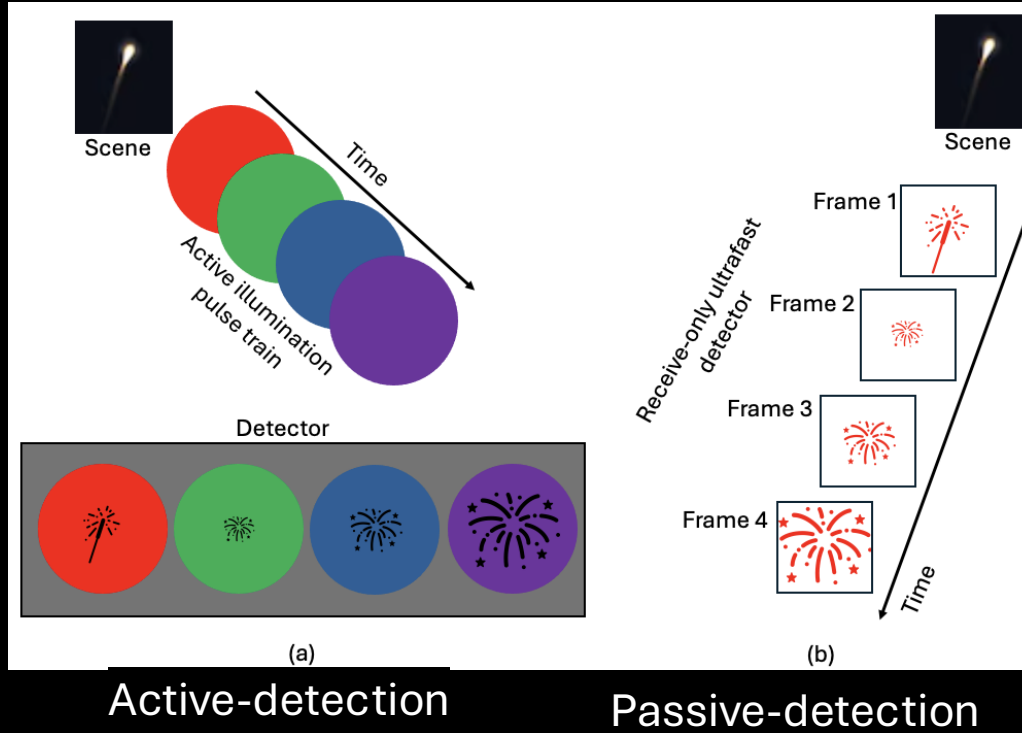
Conventional cameras capture up to 10^7

R. Kodama, P. Norreys, K. Mima, A. Dangor, R. Evans, H. Fujita, Y. Kitagawa, K. Krushelnick, T. Miyakoshi, N. Miyanaga et al., "Fast heating of ultrahigh-density plasma as a step towards laser fusion ignition," *Nature* 412, 798–802 (2001)

X. Zhu, Q. Huang, A. DiSpirito, T. Vu, Q. Rong, X. Peng, H. Sheng, X. Shen, Q. Zhou, L. Jiang et al., "Real-time whole-brain imaging of hemodynamics and oxygenation at micro-vessel resolution with ultrafast wide-field photoacoustic microscopy," *Light. Sci. & Appl.* 11, 138 (2022).

Active and Passive illumination

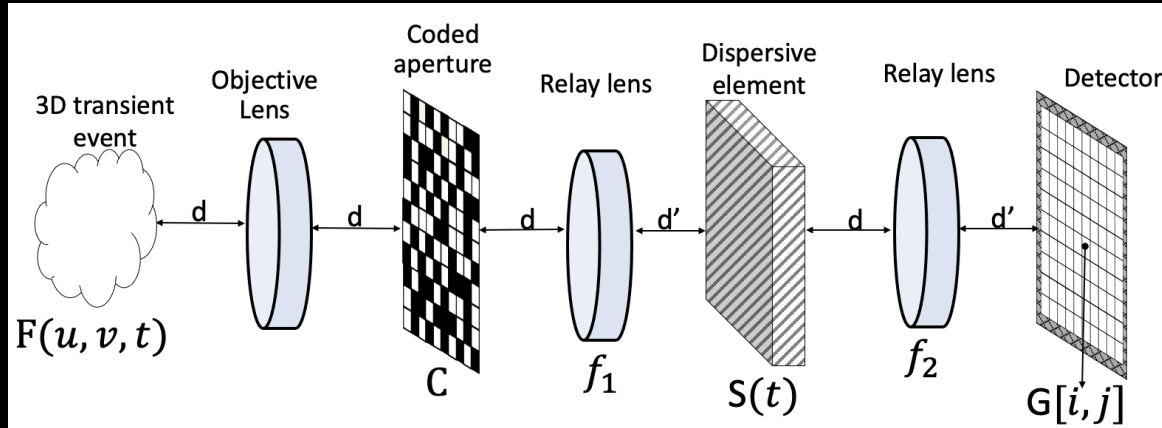
- A pulse **probe** for a transient event.
- An **optical marker** e.g. spatial division, or angle division **divide** the transmitted probe pulses
- After the transient scene in the spatial frequency domain **to recover the 3D spatio-temporal datacube**.



- Passive detection **can overcome active** illumination.
- The detector only **receive and record** the emitted and/or scattered photons from the scene.

Single snapshot compressive spectral imaging

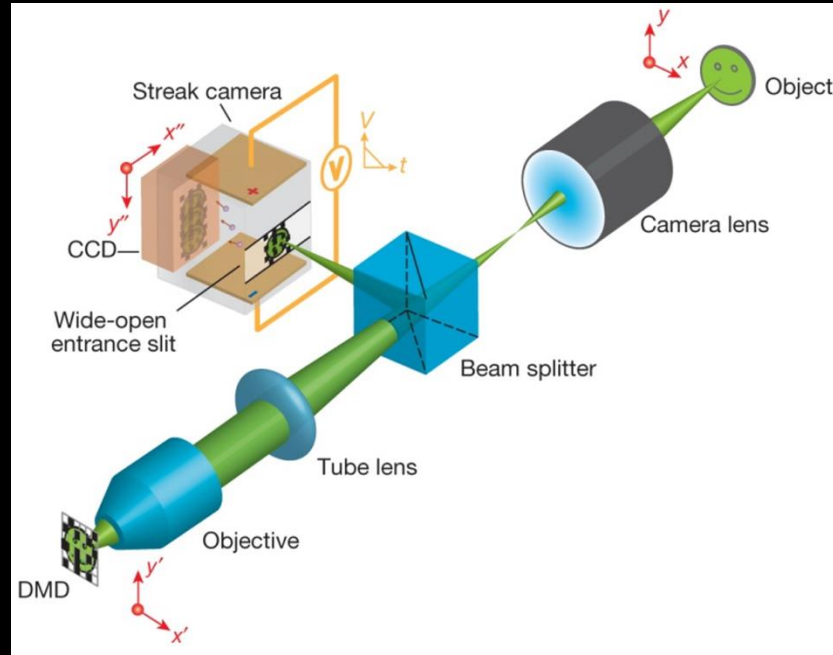
The scene is encoded using a **binary coded aperture**. Then, the encoded scene is **sheared** by the temporal **shearing operator**. The encoded and sheared scene is captured in the **detector**.



$$\hat{f} = \Psi^{-1}(\operatorname{argmin}_{\theta} \frac{1}{2} \|y - H\Psi\theta\|_2^2 + \tau \|\theta\|_1)$$

Compressive Ultrafast Photography (CUP)

The scene is encoded using a **binary coded aperture**. Then, the encoded scene is **sheared** by the temporal **shearing operator**. The encoded and sheared scene is captured in the **detector**.



- It is a **passive-detector** approach

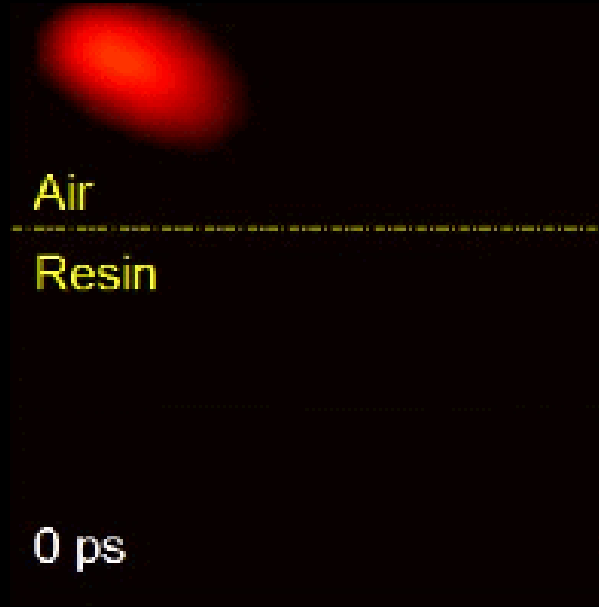
Compressive Ultrafast Photography (CUP) Applications

0 ps



Optical Mach cone using CUP

5 mm

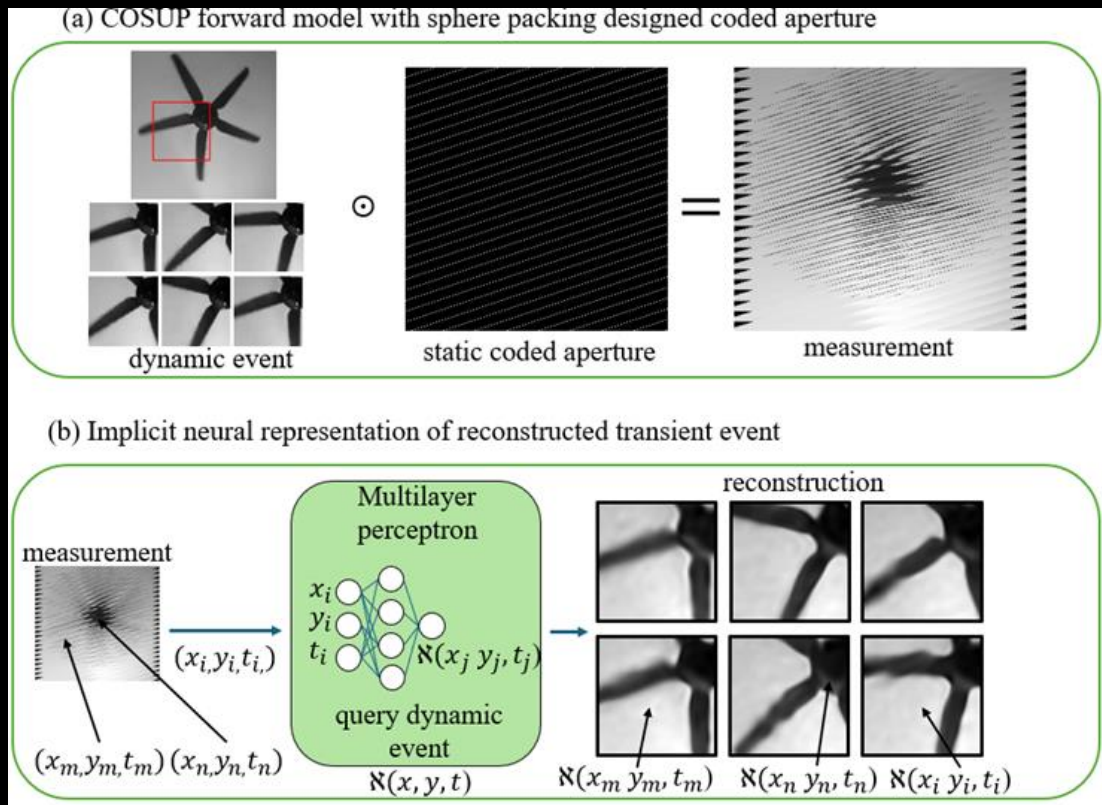


Single-shot CUP pulse refraction

Real-time applications is limited by reconstruction

L. Gao, J. Liang, C. Li, and L. V. Wang, "Single-shot compressed ultrafast photography at one hundred billion frames per second," Nature 516, 74–77 (2014).

Representación neuronal implícita para video ultrarápido

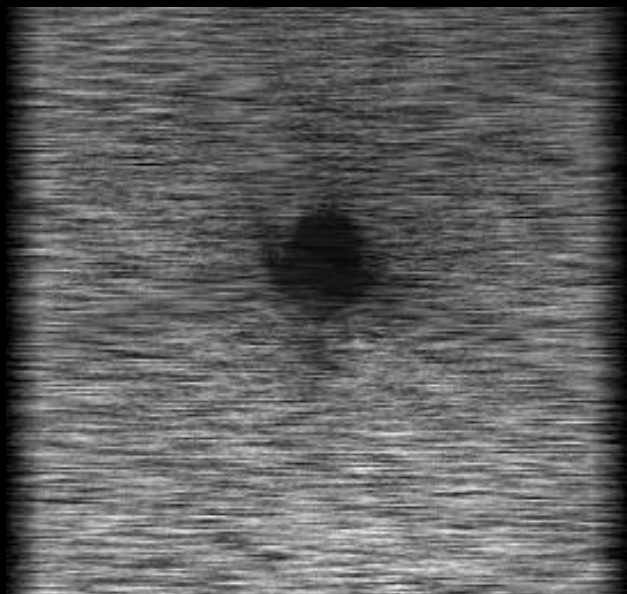


N. Díaz, M. Beniwal, M. Marquez, F. Guzmán, C. Jiang, J. Liang, and E. Vera, “Single-mask sphere-packing with implicit neural representation reconstruction for ultrahigh-speed imaging,” *Optics Express* 33, pp. 24027–24038, 2025.

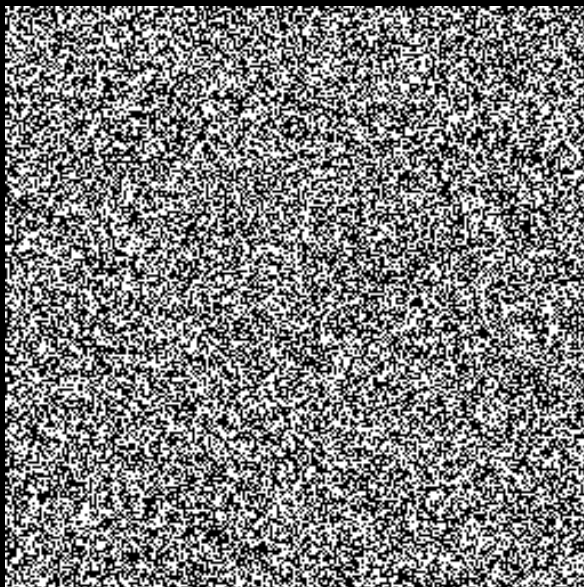
COSUP Discrete Model

$$Y_{(:,\bar{t})} = \sum_{t=0}^{T-1} X_{(:,;\bar{t})} \odot \mathcal{C}_{(:,;\bar{t})} + \Omega$$

designing the encoding is critical
for real-time applications



measurement



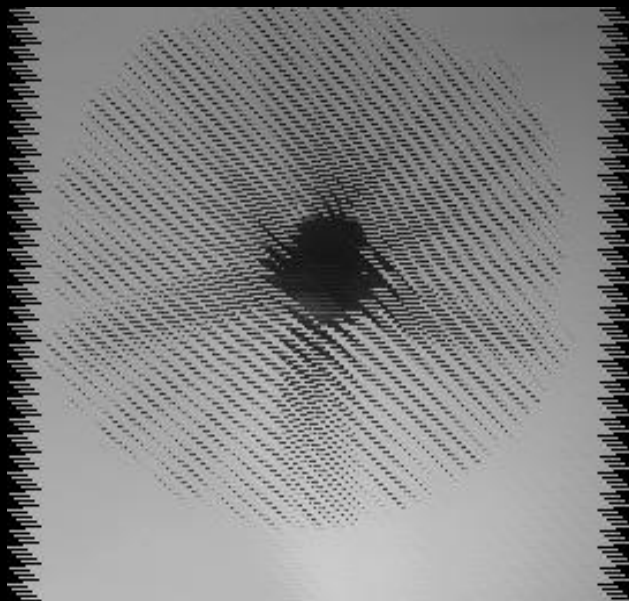
traditional coded aperture



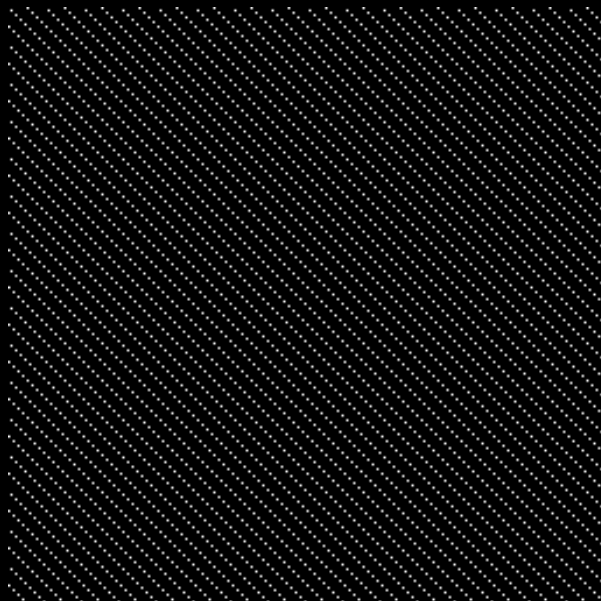
video

Discrete model

$$Y_{(:,\bar{t})} = \sum_{t=0}^{T-1} X_{(:,,\bar{t})} \odot \mathcal{C}_{(:,,\bar{t})} + \Omega \quad \text{novel aperture design}$$



measurements



coded aperture



video

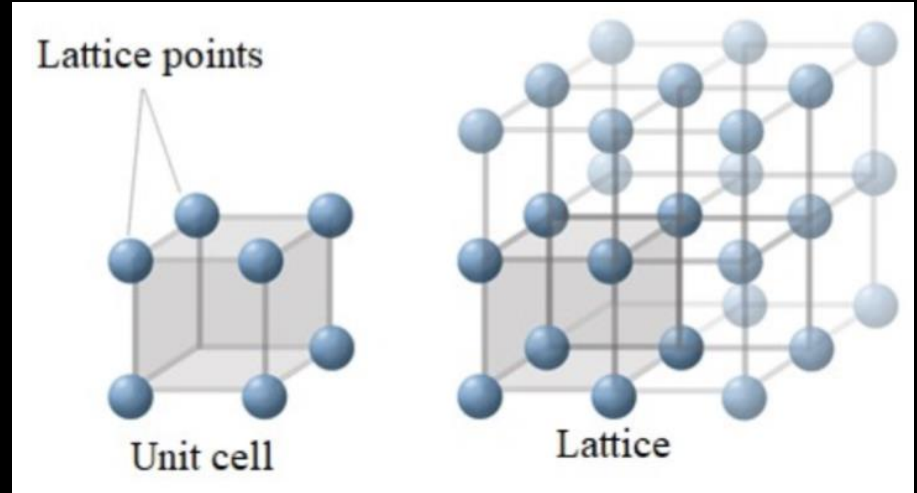
What is Sphere Packing?

- Ask for the densest packing of R^n
- Faced-center cubic lattice is optimal in 3D

$$\rho_{\Lambda_3} = \frac{\pi}{\sqrt{18}} \approx 0.74 \dots$$



stacking cannonballs



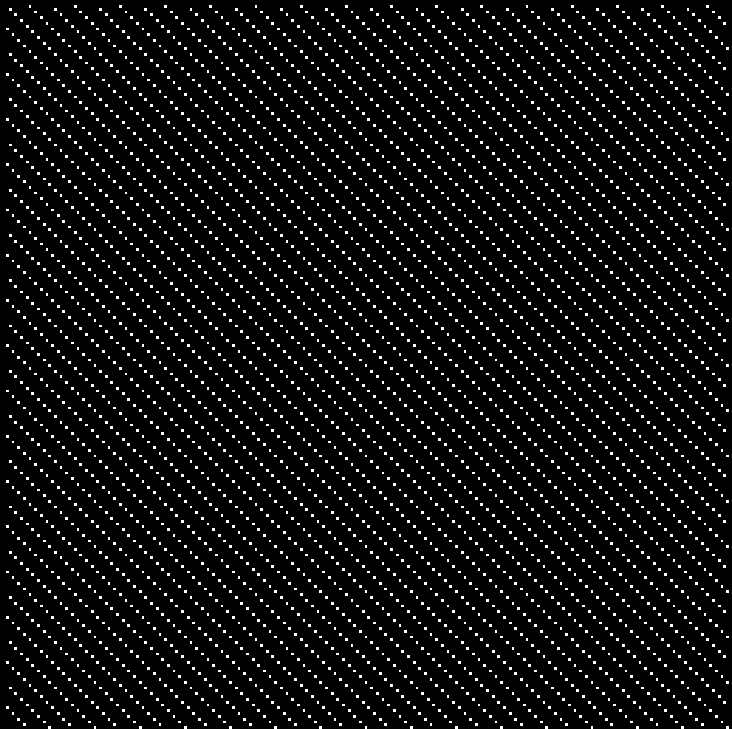
Tiling the space by replicating the unit cell

J. Kepler, The six-cornered snowflake (Paul Dry Books, 2010).

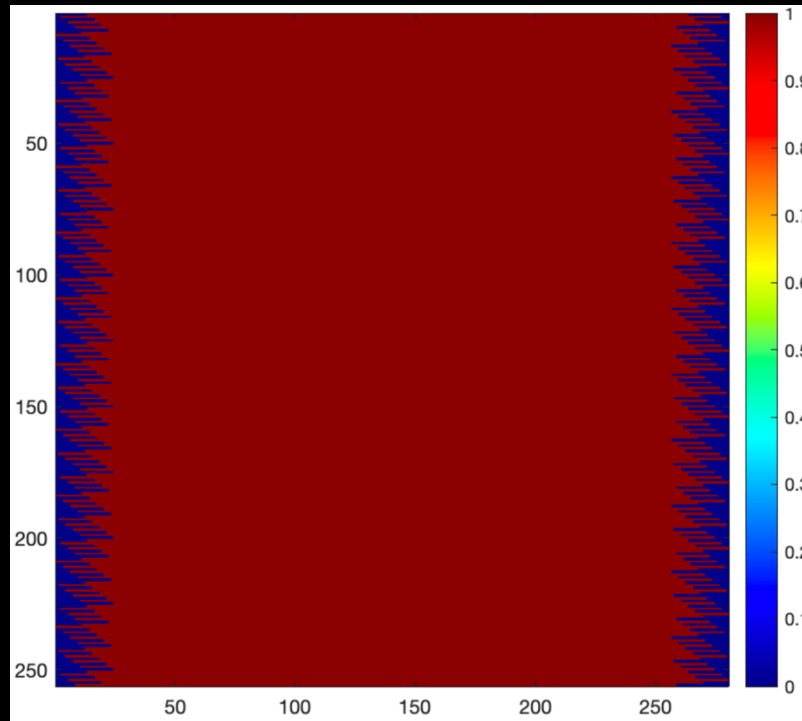
T. Hales, M. Adams, G. Bauer, T. D. Dang, J. Harrison, H. Le Truong, C. Kaliszyk, V. Magron, S. McLaughlin, T. T. Nguyen, and et al., "A formal proof of the kepler conjecture," Forum Math. Pi 5, e2 (2017).

Coded Aperture Design

- The coded aperture exploits the **shifting induced** by the galvanometer scanner.
- Our design guarantees **uniform sampling**.

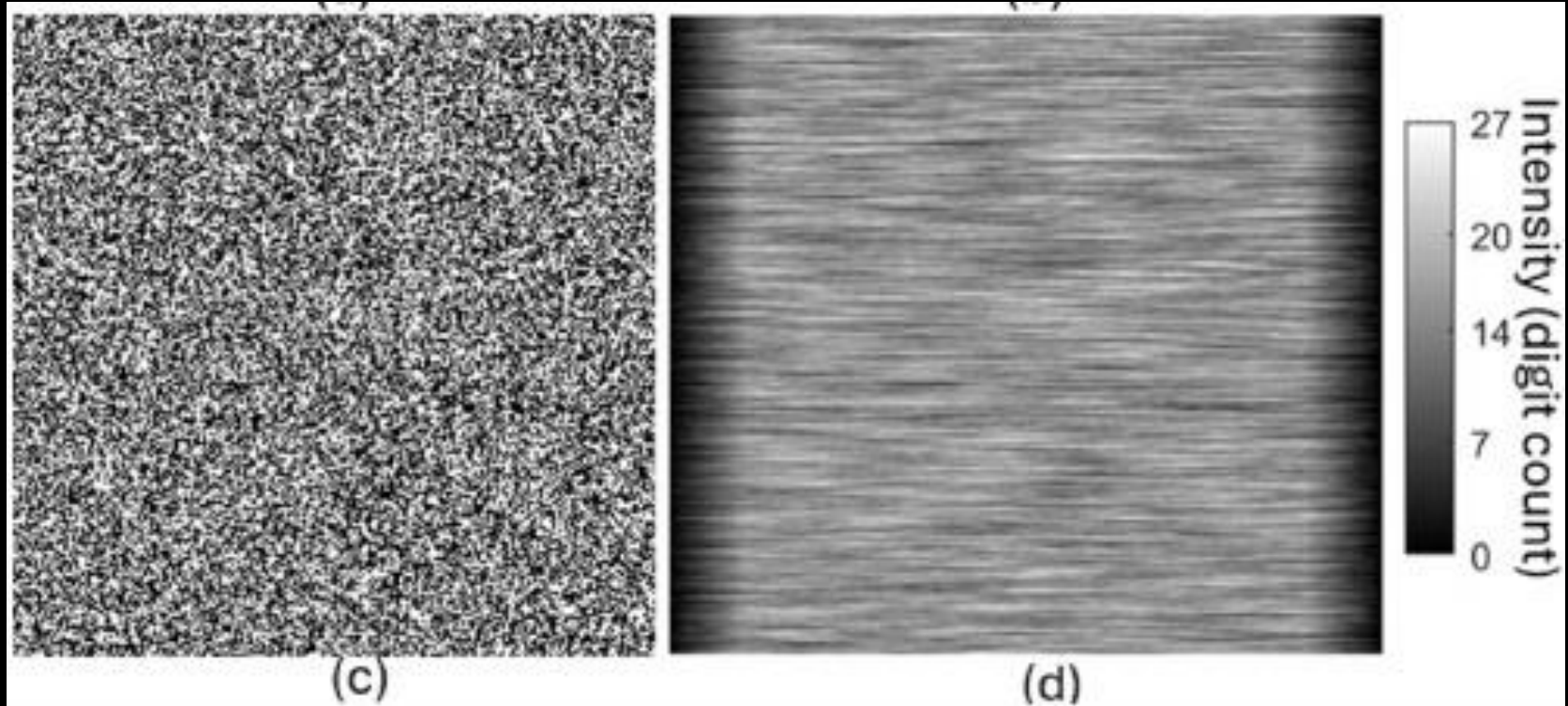


uniform coded aperture



uniform sensing

Traditional Coded Aperture Design

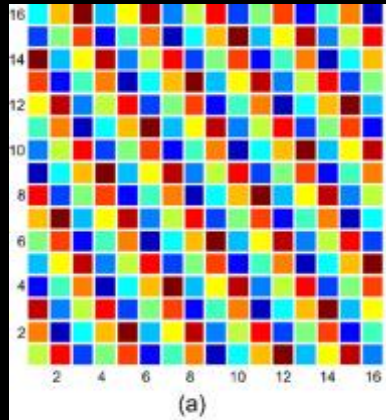


Random coded aperture

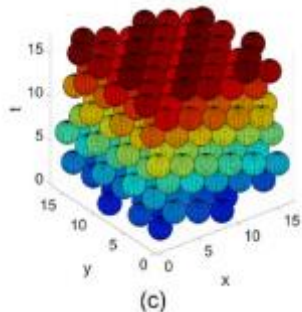
Non-uniform sensing

Sphere Packing Comparison Between DMDs

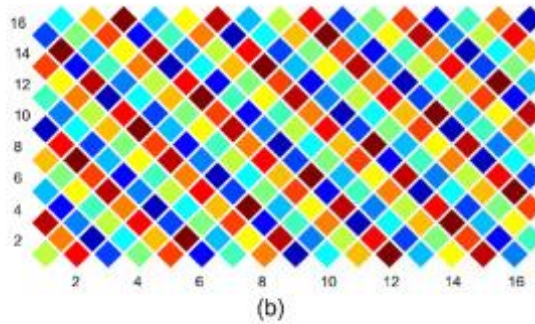
orthogonal DMD



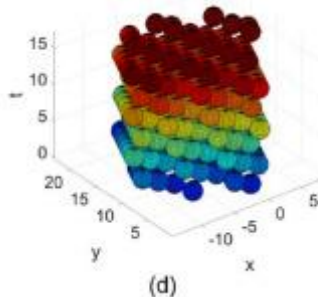
$d = 2.45$ $\rho = 0.31$



diamond DMD



$d = 2.45$ $\rho = 0.31$

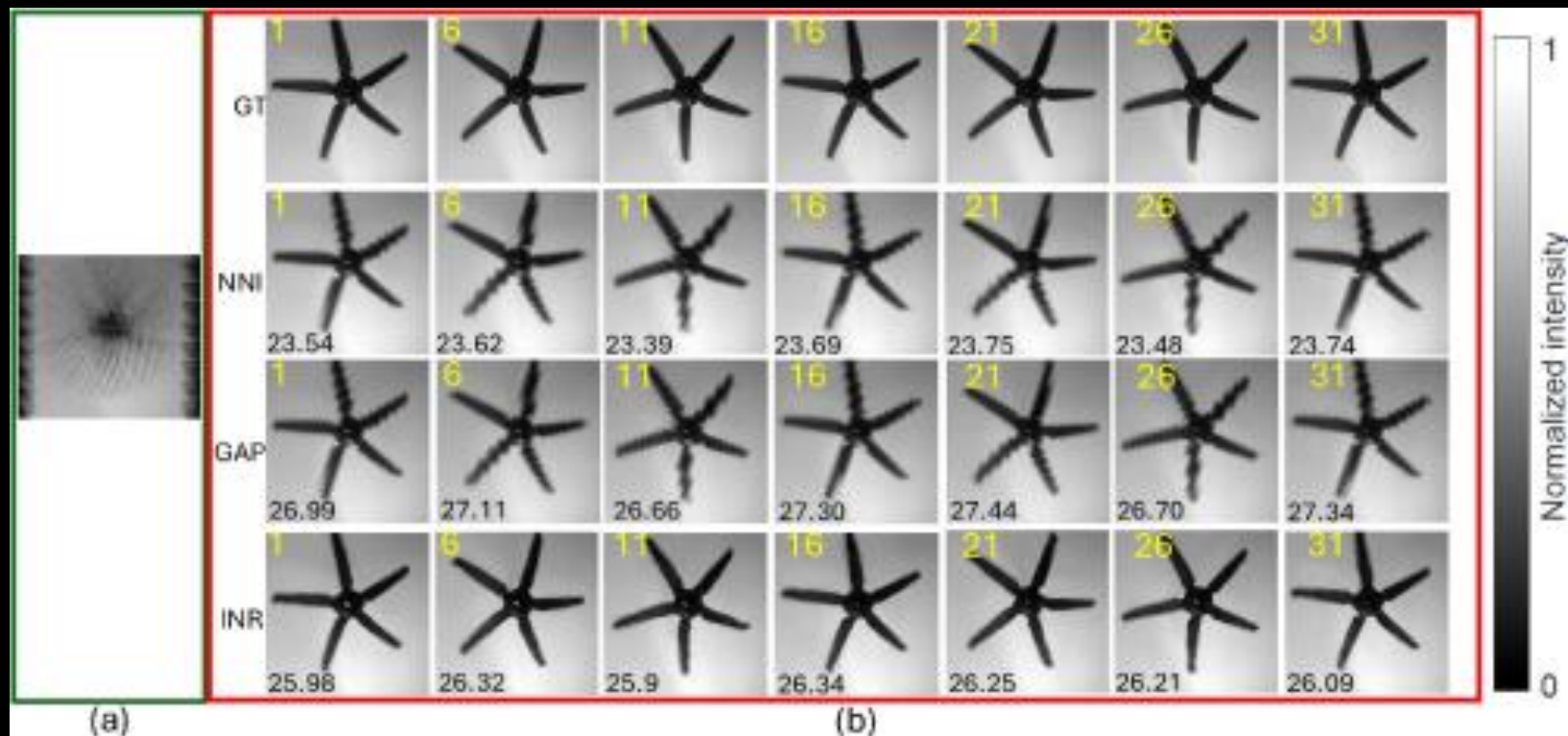


Texas instruments

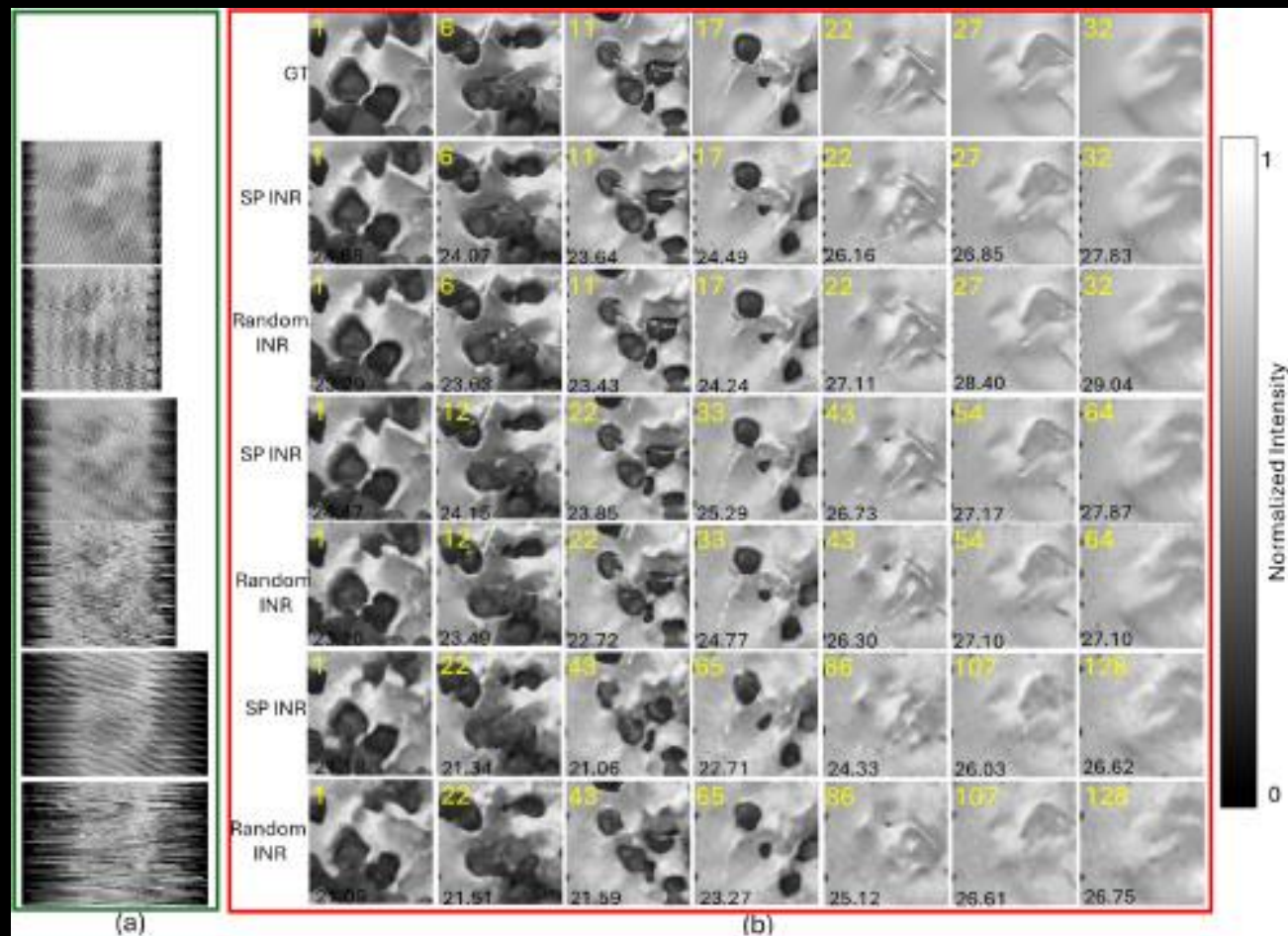
orthogonal: DLP3010LC

diamond: DLP® 0.45 WXGA DMD

Simulation Results

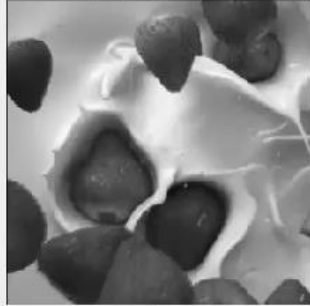


Simulation Results

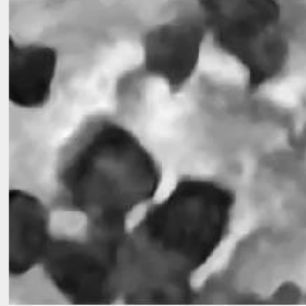


Simulation Results

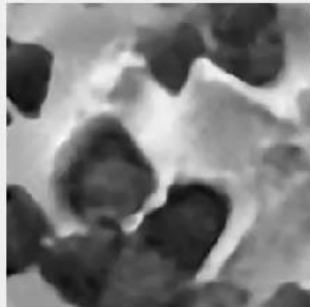
gt frame 1



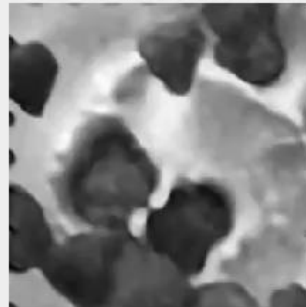
INR frame 1



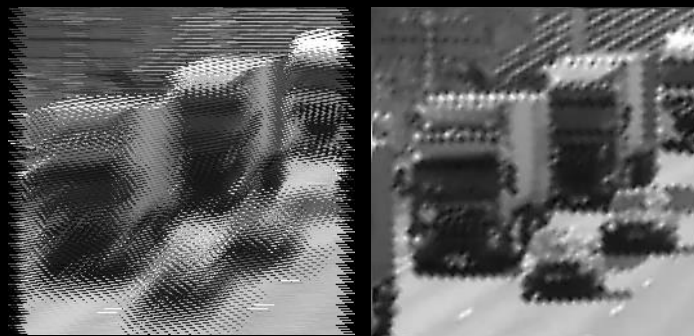
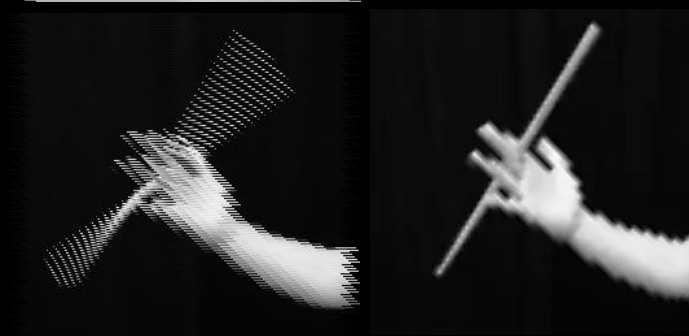
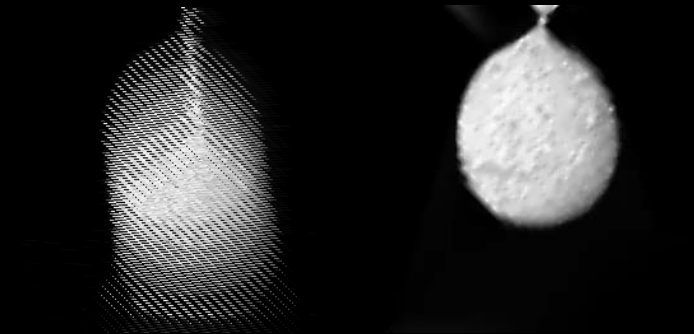
INR frame 1



INR frame 1



Simulation Results



measurements

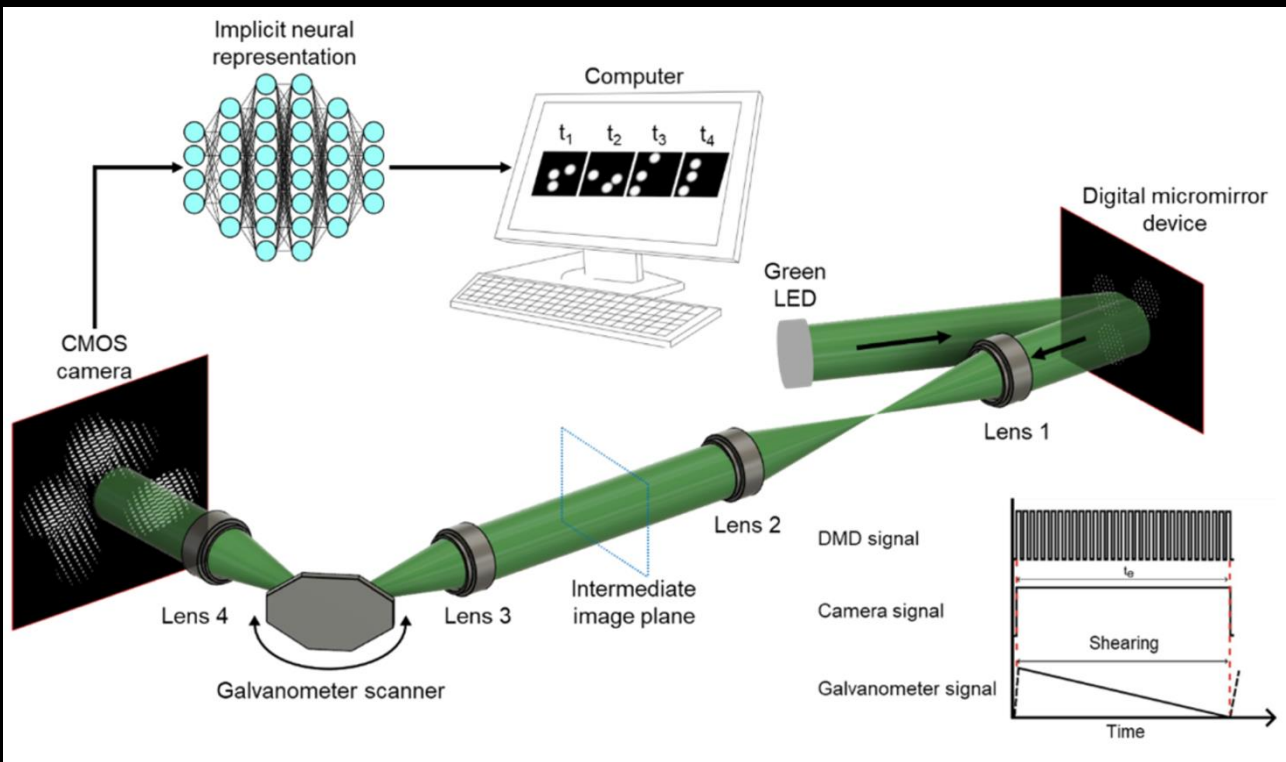
reconstruction

measurements

reconstruction

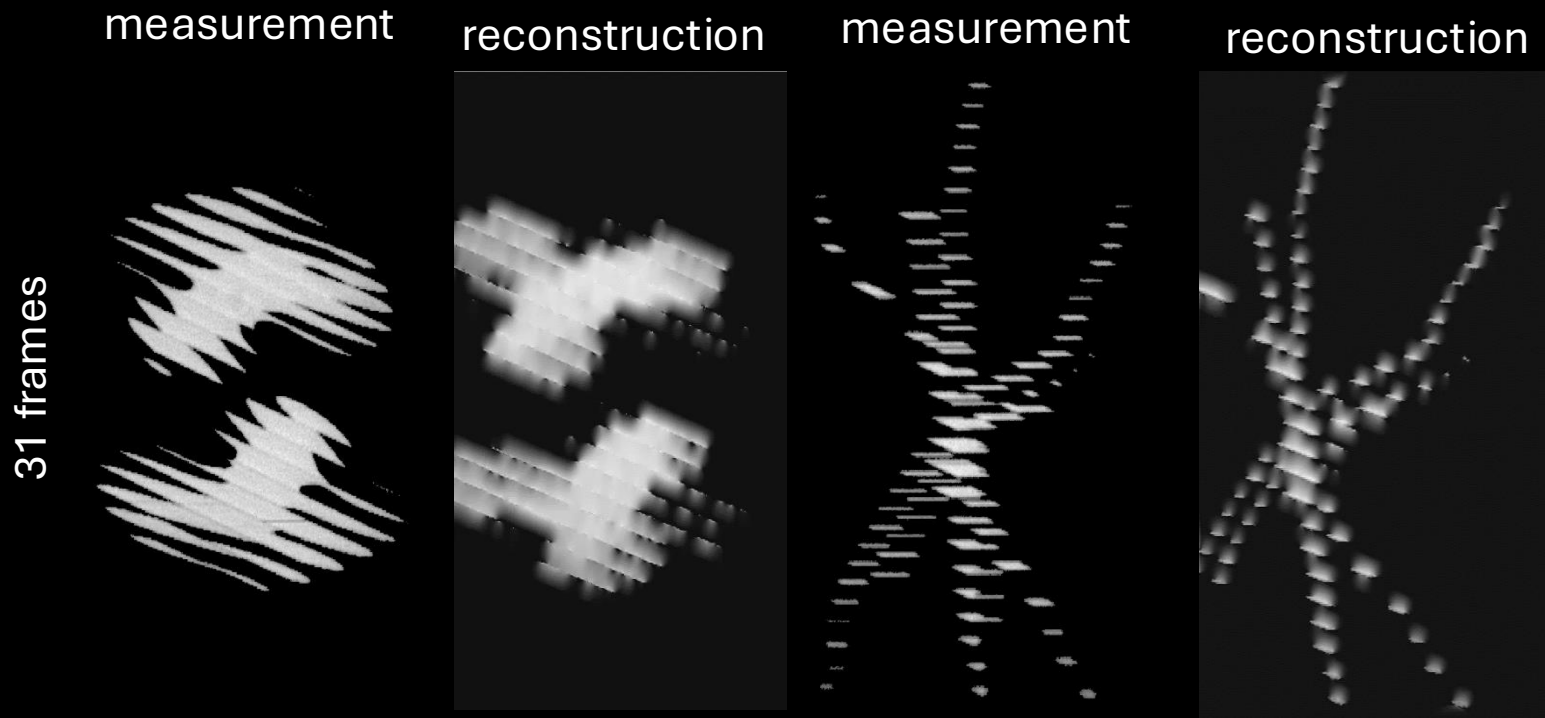
Algorithm	CA	Number of frames		
		$T = 16$	$T = 25$	$T = 31$
NNI	random CA PSNR (dB)	27.59	25.75	24.98
	random CA SSIM	0.91	0.87	0.83
	SPCA PSNR (dB)	28.33	26.9	25.06
	SPCA SSIM (dB)	0.92	0.89	0.85

Experimental implementation



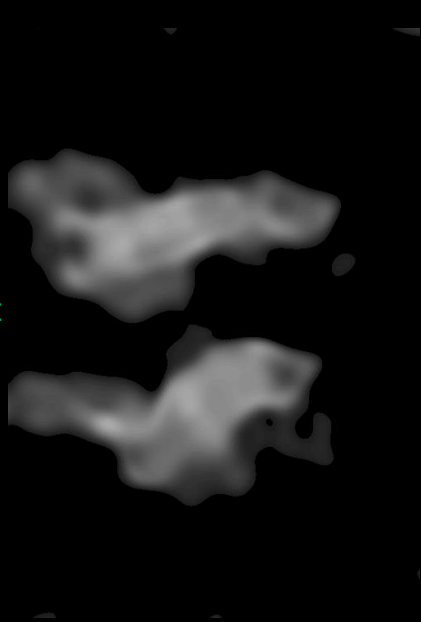
- **Architecture:** compressed optical-streaking ultra-high-speed photography
- **Encoding:** single binary mask
- **Shearing operator:** Galvanometer

Real-data: Interpolation

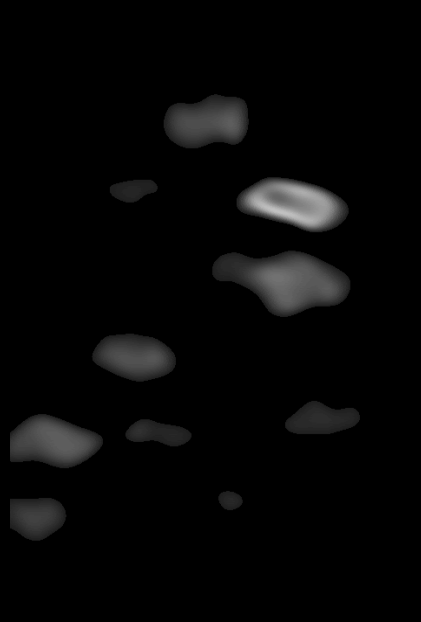


Real-data with Implicit Neural Representation

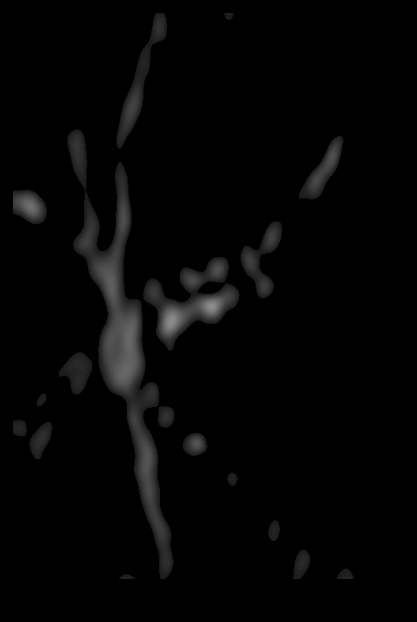
reconstruction



reconstruction



reconstruction

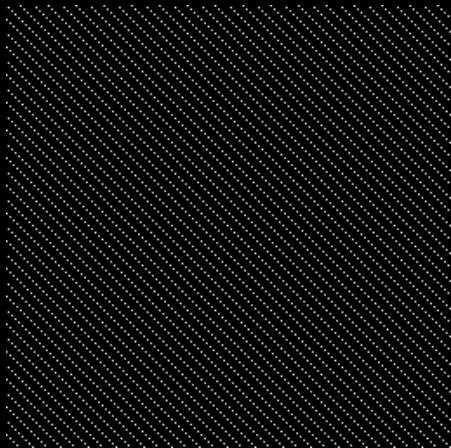


Parameters

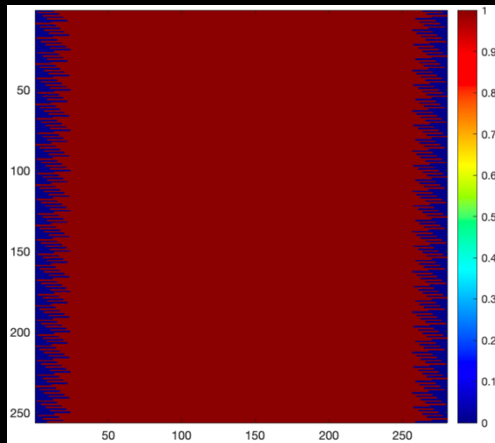
- Initial learning rate: 0.0005
- Iteration: 1500
- Resolution: 714 x 574 x 31
- Training: a single measurement
- Training time: 1.7 s
- Testing time: 2.5 ms
- Layers: Two hidden layers

Conclusions

- We introduced a novel **binary coded aperture** for compressed **optical-streaking** ultra-high-speed photography
- Our designed coded aperture samples uniformly the video
- We recover the underlying video using interpolation and **implicit neural representation**



uniform coded aperture



uniform sensing

